# Lesson 4.1: Data modelling, star schemas, DAX and publishing

## What you learn

- **Star schema.** A central fact table (sales) linked to dimension tables (date, product, customer).
- **DAX.** CALCULATE changes the filter context of a measure. SAMEPERIODLASTYEAR returns the same dates in the previous year, so you can compare with it.
- **Publishing.** Publish to the Power BI Service, set refresh and share by workspace with correct access.

## Tools and resources (official sites, not checked live)

- Power BI documentation

ASSIGNMENT

### Practice for lesson 4.1

**Business problem:** Write a year-on-year measure.

**Steps**

1. Create Total Sales = SUM(Sales[Amount]).
2. Create Sales LY = CALCULATE([Total Sales], SAMEPERIODLASTYEAR('Date'[Date])).
3. Create the growth percentage.

**Expected output:** A card showing growth versus last year.

SOLUTION (ANSWER KEY)

Sales LY needs a proper date table marked as the date table. Growth % = DIVIDE([Total Sales] - [Sales LY], [Sales LY]).

## Check yourself

1. What is a fact table?
   *Answer: The central table of events such as sales.*
2. What does SAMEPERIODLASTYEAR need?
   *Answer: A continuous date table.*

Lesson 4.1 of the **Data Analytics and Data Science** lessons. See this lesson and its official links on skillnavora.in · All chapters and lessons · Open the full lesson guide. Tools, prices and plans change; confirm on each official site.